Aws

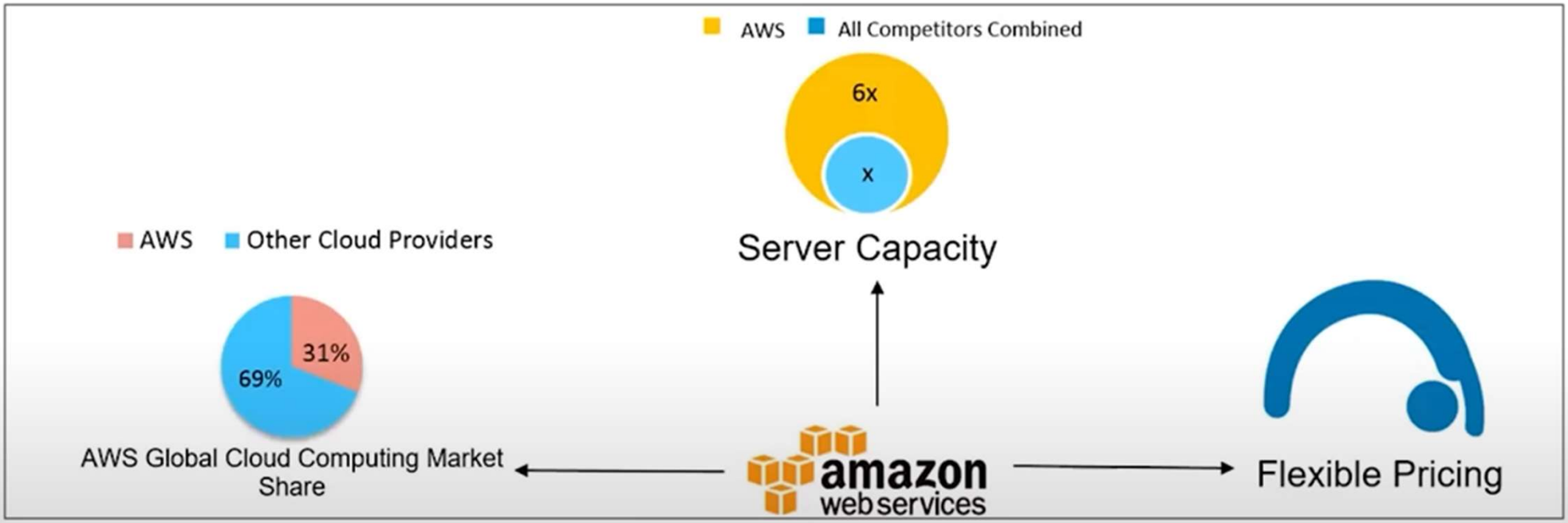
Amazon Web Services

Dr Mukti Padhya

Assistant Professor

SCSDF, NFSU

Why AWS


AWS
All Competitors Combined
6x
x
Server Capacity
AWS
Other Cloud Providers
31%
69%
AWS Global Cloud Computing Market Share
amazon web services
Flexible Pricing

Amazon Web Service (AWS)

Amazon Web Services (AWS) is a bundled remote computing service that provides cloud computing infrastructure over the Internet with:

- Storage
- Bandwidth
- Customized support for application programming interfaces (API)



Who Is Using AWS?



Benefits Of AWS



Service Comparison: AWS, Azure And GCP

Amazon AWS	Microsoft Azure	GCP
S3	Blob Storage	Storage
EC2	Virtual Machines	Compute Engine
EC2 Container Service	Container Service	Kubernetes Engine
Elastic Beanstalk	Azure App service	App Engine
DynamoDB	Cosmos DB	Cloud Datastore
RDS	SQL Database	BigQuery
Lambda	Azure Functions	Cloud Functions

AWS Global Infrastructure



Regions and Number of Availability Zones

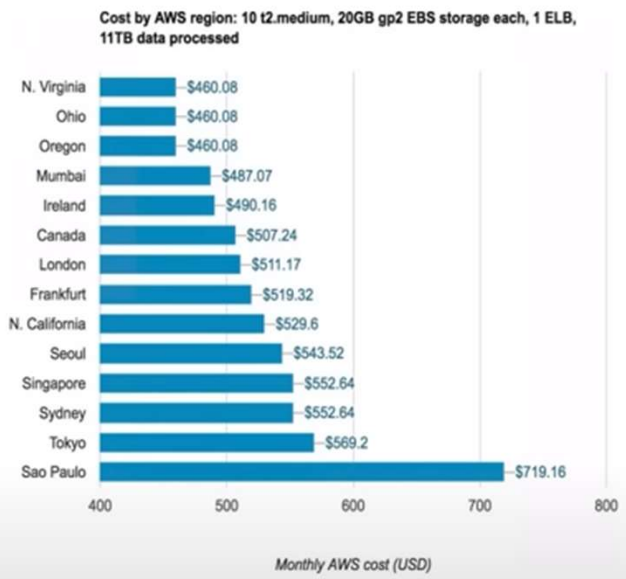
New Regions (Coming Soon)

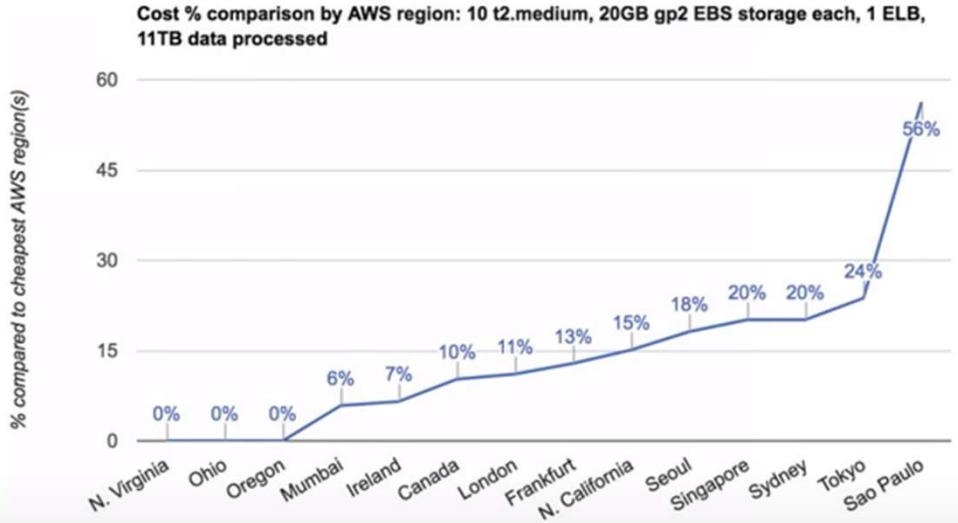
Example: Mumbai is a Region with 2 Availability Zones

How To Choose A Right Region?



How To Choose A Right Region?





Sao Paul is 56% costlier than N. Virginia
N Virginia is the cheapest one among all

What Is Availability In AWS?

The term ***Availability*** refers to the ability of a system or component to be operational and accessible anytime

It ensures that data and service is available in normal and even in disaster recovery operations

It always answers the question "***Can I access my data right now***"

It is achieved by maintaining reliable engineering, service management techniques and redundancy so as to minimize system failures and time to repair

Considering, all these aspects AWS provides optimum services to its customers via ***High Availability***



AWS Availability Zones (AZs)

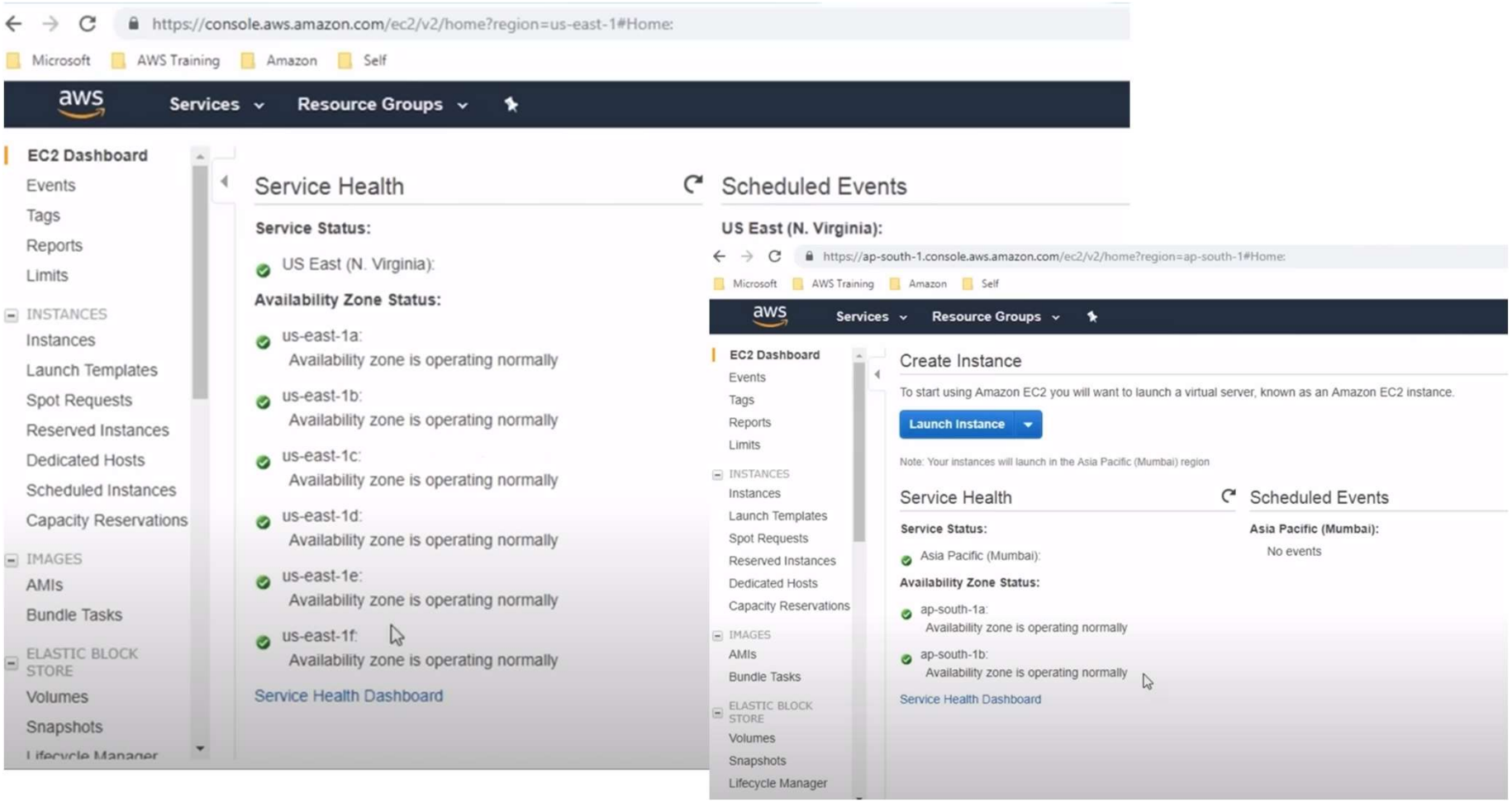
01 ***Availability Zones*** are locations carrying data centers in a ***Region*** which are used by customers and each Region has ***two or more AZs***

02 AWS Availability Zones offer their customers a simple and more effective way to design and operate applications and databases



03 Availability Zones make these applications and databases more highly available, fault tolerant and scalable then traditional data centres

Note: AWS Global Infrastructure has 60 Availability Zones


https://console.aws.amazon.com/ec2/v2/home?region=us-east-1#Home:
Microsoft
AWS Training
Amazon
Self
aws
Services
Resource Groups
EC2 Dashboard
Events
Tags
Reports
Limits
INSTANCES
Instances
Launch Templates
Spot Requests
Reserved Instances
Dedicated Hosts
Scheduled Instances
Capacity Reservations
IMAGES
AMIs
Bundle Tasks
ELASTIC BLOCK STORE
Volumes
Snapshots
Lifecycle Manager
Service Health
Service Status:
US East (N. Virginia):
Availability Zone Status:
us-east-1a:
Availability zone is operating normally
us-east-1b:
Availability zone is operating normally
us-east-1c:
Availability zone is operating normally
us-east-1d:
Availability zone is operating normally
us-east-1e:
Availability zone is operating normally
us-east-1f:
Availability zone is operating normally
Service Health Dashboard
Scheduled Events
US East (N. Virginia):
https://ap-south-1.console.aws.amazon.com/ec2/v2/home?region=ap-south-1#Home:
Microsoft
AWS Training
Amazon
Self
aws
Services
Resource Groups
EC2 Dashboard
Events
Tags
Reports
Limits
INSTANCES
Instances
Launch Templates
Spot Requests
Reserved Instances
Dedicated Hosts
Capacity Reservations
IMAGES
AMIs
Bundle Tasks
ELASTIC BLOCK STORE
Volumes
Snapshots
Lifecycle Manager
Create Instance
To start using Amazon EC2 you will want to launch a virtual server, known as an Amazon EC2 instance.
Launch Instance
Note: Your instances will launch in the Asia Pacific (Mumbai) region
Service Health
Service Status:
Asia Pacific (Mumbai):
Availability Zone Status:
ap-south-1a:
Availability zone is operating normally
ap-south-1b:
Availability zone is operating normally
Service Health Dashboard
Scheduled Events
Asia Pacific (Mumbai):
No events

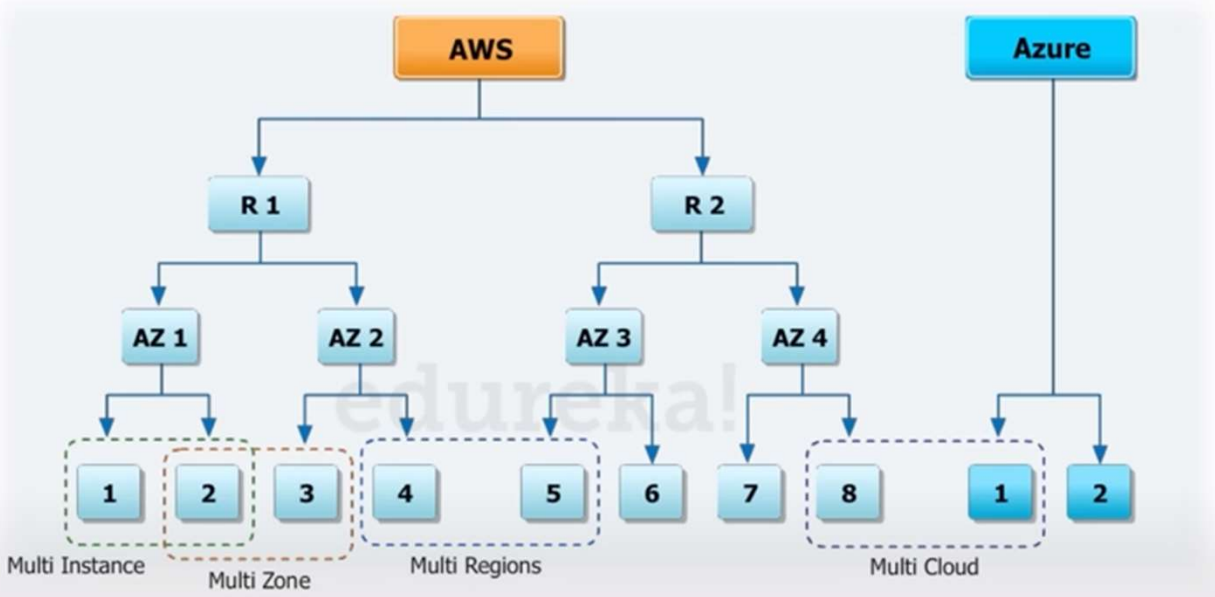
What Is High Availability (HA)?

High Availability refers to a system or component that is continuously operational for a desirably long duration oftime

- In order to get High availability users deploy their servers in two ***Availability Zones***, so that in case one AZ goes down then still the application remains on by using other AZ
- High Availability ensures ***fault tolerance*** and ***low latency*** in a particular Region



Ways To Implement High Availability



Availability

Lower — Price — Higher

In case of launching/planning a second instance in a new region the instance price will differ and also an additional charges for region data transfer will be added

How To Measure Availability?

- Downtime Per Year (Hours) = (1– Uptime Ratio) x 365 x 24
- Where Uptime ratio or Availability = MTBF/(MTBF+MTTR)
- MTBF : Mean Time Between Failures
- MTTR : Mean Time To Repair
- MTBF = Hours in Year / No. of Failure
- MTTR = (No. of Device x Time to Repair the Device) / Total No. of Failures

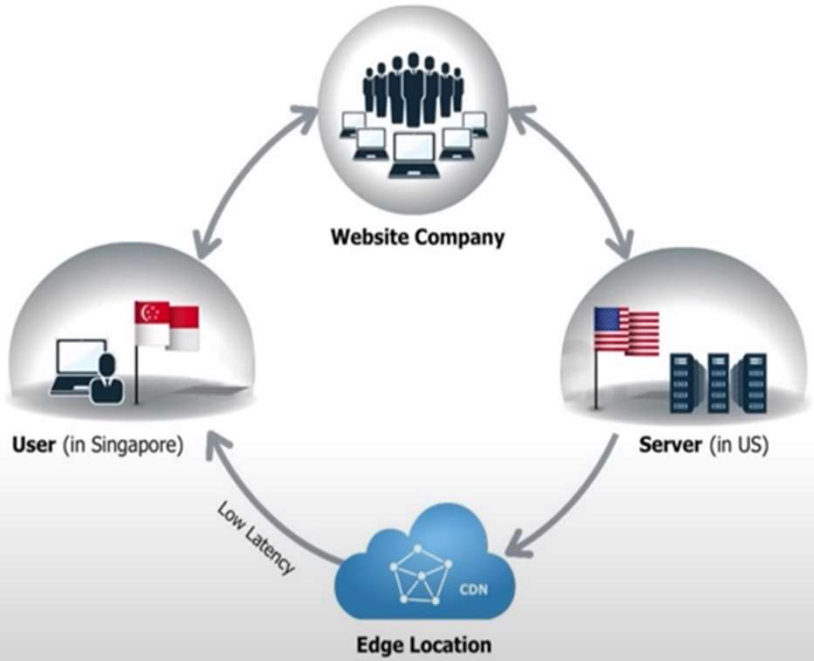
Calculating the availability of a system -

A = (Uptime / Uptime + Downtime) x 100% = X%

Where: A = the percentage of time that the system is operational

What Is Edge Location?

- ***Edge Location*** is an intermediate between the end users and servers to access the services from AWS
- It is a small setup in different location to provide low latency connection by caching static content
- Basically, it's a Content Delivery Network and used with AWS CloudFront Service



Demo – Signing Up For Free Tier Account With AWS

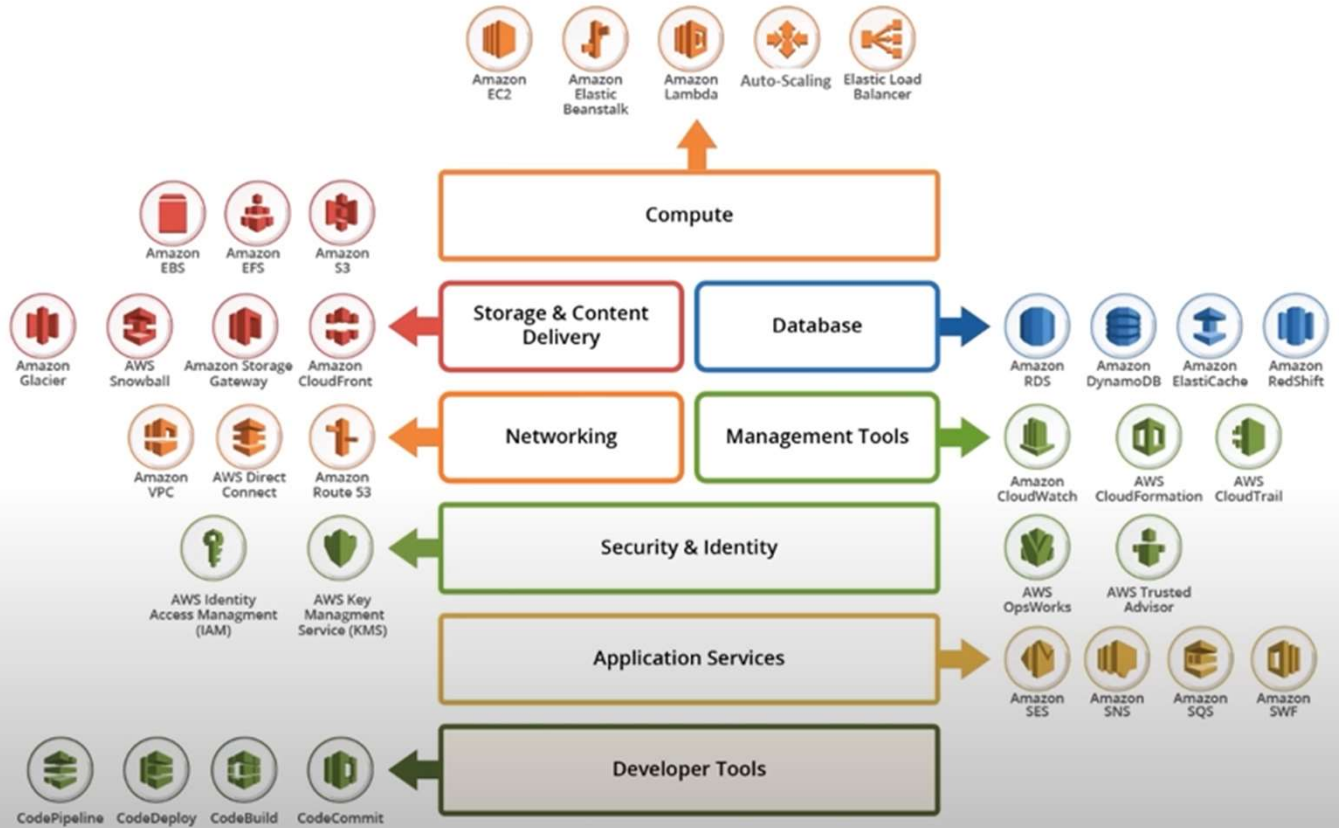
1. AWS provides 12 months free account including all basic services like S3, EC2, DynamoDB etc.
2. Open https://portal.aws.amazon.com/billing/signup#/start and fill basic details to create an account
3. Click on continue and Select account type as personal
4. Fill all mandatory fields and click on proceed
5. Proceed to payment gateway to map your credit card
6. Don't panic, only 2 Rupees will get deducted from your account and you will get all basic services free for 12 months

What Do You Get In Free Tier Account?

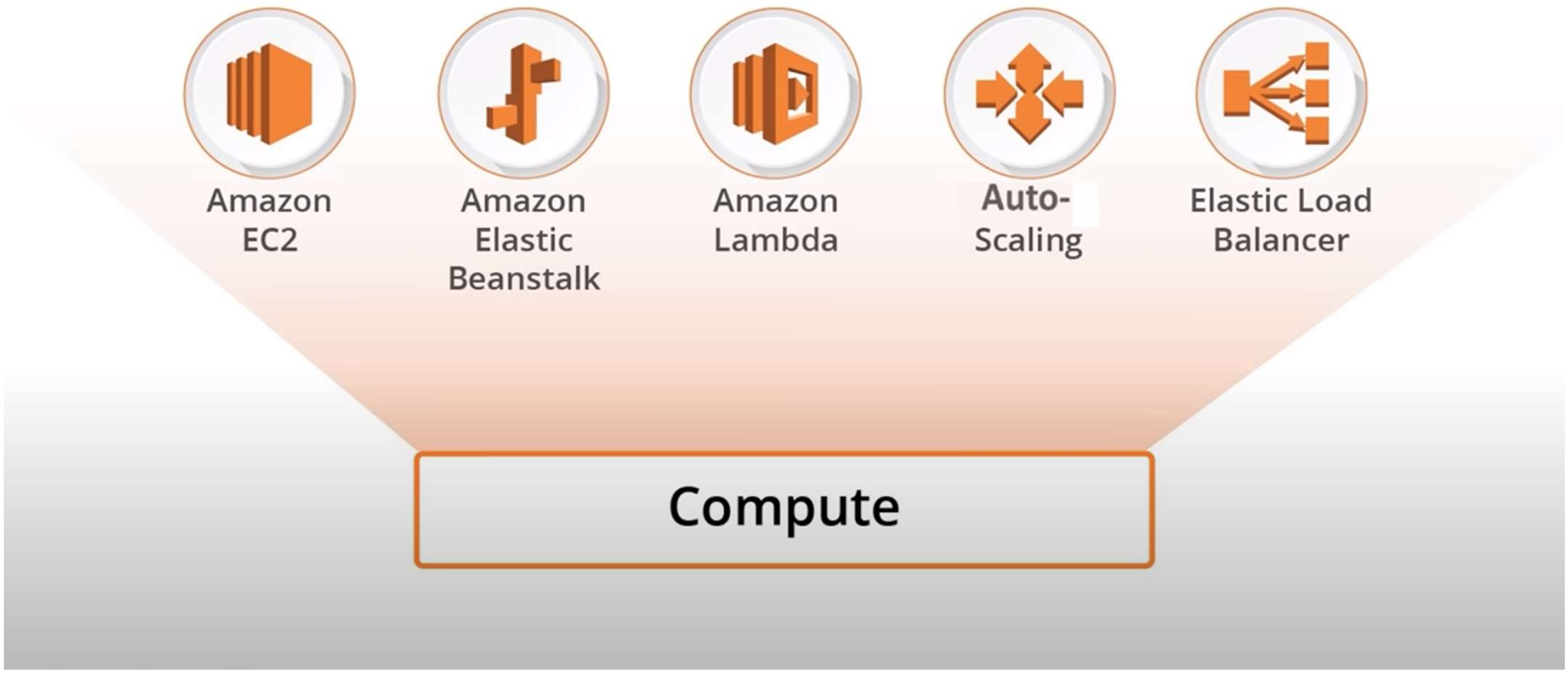
Services	Benefits(1 Year)
Amazon Elastic Cloud Compute 	» 750 hours/month of Linux, RHEL or SLES t2.micro instance usage » 750 hours/month of Windows t2.micro instance usage
Elastic Load Balancer 	» 750 hours + 15 GB Data Processing
Elastic Block Storage 	» 30GB in combination of SSD/Magnetic + 2 million I/O and 1 GB of snapshot storage
Amazon Web Services 	» 15GB of Bandwidth aggregated across all Services » 1GB of Regional Data Transfer

Free-Tier Account provides all the necessary Services for practicing the hands on for AWS Architect

AWS Services



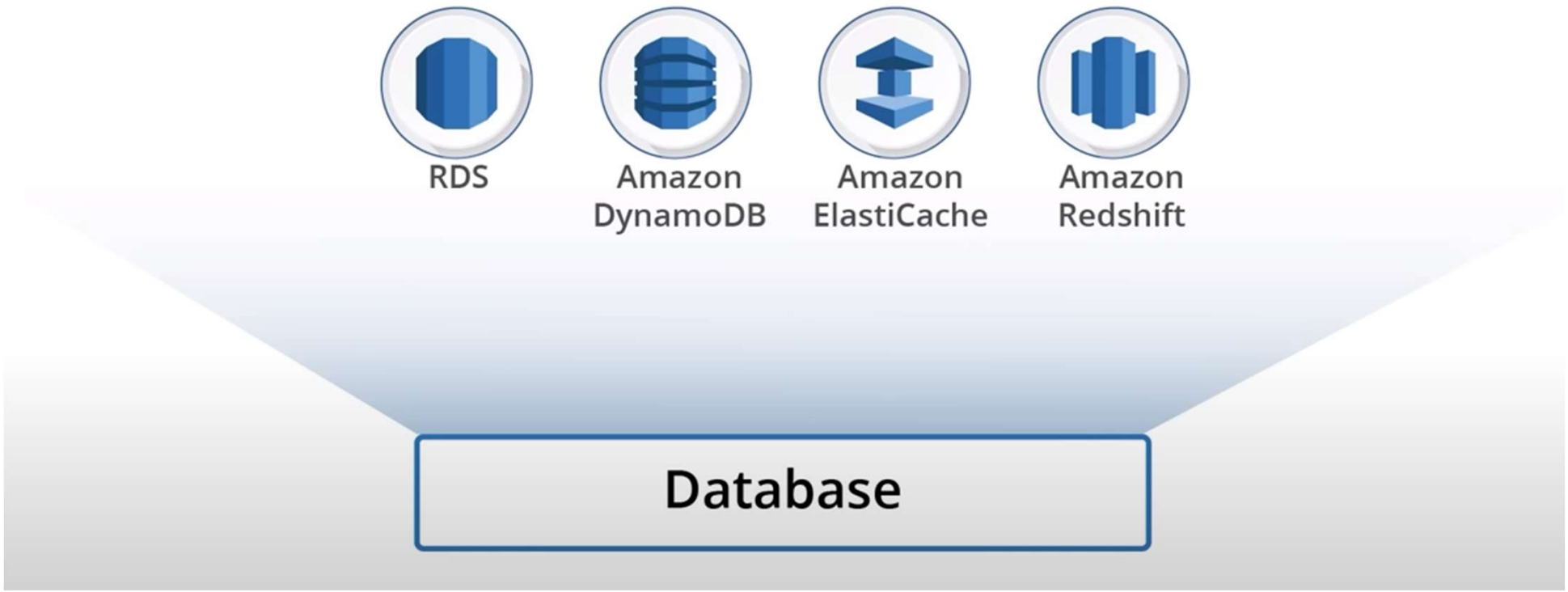
Compute Services



Storage And Content Delivery Services



Database Services



Management Tools

